

Two positive results for the measurable polydisc Douglas–Rudin problem

Candidate partial-result packet

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Status. Candidate substantial partial result; likely valid; human review recommended. The essential-supremum problem for arbitrary measurable functions remains open here.

1 Source question

Kumar, Rastogi, and Tripathi ask whether the Douglas–Rudin theorem extends to measurable unimodular functions on the distinguished boundary $\mathbb{T}^d$ of the polydisc [3]. McDonald proved the uniform theorem for *continuous* unimodular functions, with rational inner factors [2]. The source observes that a measurable scalar theorem would also feed into its operator-valued method. Its concluding formulation (PDF page 10) is reproduced below.

operator-valued inner functions [11, 12], yet a comprehensive understanding remains elusive. We believe that our generalization of the Douglas–Rudin approximation theorem holds promise in resolving Douglas–Rudin type factorization results for the operator-valued setting.

We end with a natural remark about a possible extension of our results. It is natural to wonder if the Douglas–Rudin theorem holds for (operator-valued) unimodular functions defined on $\mathbb{T}^d$. Clearly, our proof technique would yield the result, if the Douglas–Rudin theorem for scalar-valued unimodular functions were available for $\mathbb{T}^d$. However, to the best of our knowledge, the result available in this direction is only available for the continuous unimodular functions [22]. It would be interesting to explore the possibility of extending the result in [22] to measurable unimodular function—which would then yield the result in operator-valued case as well following our technique.

REFERENCES

- [1] A. B. Aleksandrov, *Existence of inner functions in the unit ball*, Mat. Sb. **118** (1982), no. (160), N2(6), 147–163.
- [2] D. Alpay, T. Bhattacharyya, A. Jindal, and P. Kumar, *A dilation theoretic approach to approximation by inner functions*, Bulletin of the London Mathematical Society **55** (2023), no. 6, 2840–2855.
- [3] S. Axler, *Factorization of L^∞ functions*, Annals of Mathematics **106** (1977), no. 3, 567–572.

Write m_d for normalized Haar measure on $\mathbb{T}^d$. If Φ, Ψ are polydisc inner functions, their boundary quotient means $\Phi\Psi^{-1} = \Phi\overline{\Psi}$ almost everywhere.

2 Idea of the proofs

There are two different ways to retain a complete theorem without claiming the open uniform statement.

First, Haar regularity lets us replace a measurable phase, outside a set of arbitrarily small measure, by a continuous phase. McDonald’s theorem then gives a rational-inner quotient uniformly close to that continuous phase. This proves density in every finite L^r and, after choosing summable errors, almost-everywhere convergence.

Second, suppose all measurable dependence is carried by finitely many analytic characters $\zeta \mapsto \zeta^a$. Apply the one-variable Douglas–Rudin theorem to each one-variable phase and compose its inner factors with $z \mapsto z^a$. Those compositions are polydisc inner, and multiplication assembles the factors without any exceptional set. This gives genuine essential-supremum approximation for a broad class containing all coordinate-separable phases.

3 Finite- L^r density for every measurable phase

Let $\mathcal{Q}_{\text{rat}}(\mathbb{T}^d)$ denote the boundary quotients $\Phi\bar{\Psi}$ with Φ, Ψ rational inner on $\mathbb{D}^d$.

Theorem 1 (finite-norm Douglas–Rudin approximation). *Let $d \geq 1$, let $0 < r < \infty$, and let $f : \mathbb{T}^d \rightarrow \mathbb{T}$ be measurable. For every $\varepsilon > 0$ there is $q \in \mathcal{Q}_{\text{rat}}(\mathbb{T}^d)$ such that*

$$\|f - q\|_{L^r(m_d)} < \varepsilon.$$

Moreover, there is a sequence $(q_n) \subset \mathcal{Q}_{\text{rat}}(\mathbb{T}^d)$ such that $q_n(\zeta) \rightarrow f(\zeta)$ for almost every $\zeta \in \mathbb{T}^d$.

Proof. Fix small $\eta, \delta > 0$. Choose a finite η -net $\{\lambda_1, \dots, \lambda_N\} \subset \mathbb{T}$ and a measurable partition $\mathbb{T}^d = E_1 \sqcup \dots \sqcup E_N$ such that the simple phase $s = \sum_j \lambda_j \mathbf{1}_{E_j}$ satisfies $|f - s| < \eta$ everywhere after changing representatives on a null set.

By inner regularity, choose compact $K_j \subset E_j$ with

$$m_d\left(\mathbb{T}^d \setminus \bigcup_j K_j\right) < \delta.$$

The finitely many disjoint compact sets have pairwise disjoint open neighborhoods U_j . Choose continuous $0 \leq \rho_j \leq 1$ supported in U_j and equal to 1 on K_j . Writing $\lambda_j = e^{i\theta_j}$, define

$$c(\zeta) = \exp\left(i \sum_j \theta_j \rho_j(\zeta)\right).$$

Then c is continuous and unimodular, and $c = \lambda_j$ on K_j .

McDonald’s polydisc theorem gives $q \in \mathcal{Q}_{\text{rat}}(\mathbb{T}^d)$ with $\|c - q\|_\infty < \eta$. On $K = \bigcup_j K_j$ we have $|f - q| < 2\eta$, while everywhere $|f - q| \leq 2$. Hence

$$\int_{\mathbb{T}^d} |f - q|^r dm_d \leq (2\eta)^r + 2^r \delta.$$

Taking η and δ sufficiently small proves density for every finite r , including $0 < r < 1$.

For the final assertion, apply the proved case $r = 1$ with $\|f - q_n\|_1 < 2^{-n}$. Tonelli’s theorem gives $\sum_n |f - q_n| < \infty$ almost everywhere, and therefore $q_n \rightarrow f$ almost everywhere. $\square$

4 A uniform measurable subclass

For $a = (a_1, \dots, a_d) \in \mathbb{N}_0^d$, put $z^a = z_1^{a_1} \dots z_d^{a_d}$ and $|a| = \sum_\ell a_\ell$.

Theorem 2 (analytic-character pullbacks). *Let $a_1, \dots, a_N \in \mathbb{N}_0^d \setminus \{0\}$ and let $g_1, \dots, g_N : \mathbb{T} \rightarrow \mathbb{T}$ be arbitrary measurable functions. Define*

$$f(\zeta) = \prod_{j=1}^N g_j(\zeta^{a_j}) \quad (\zeta \in \mathbb{T}^d).$$

For every $\varepsilon > 0$ there are inner functions Φ, Ψ on $\mathbb{D}^d$ such that

$$\operatorname{ess\,sup}_{\zeta \in \mathbb{T}^d} |f(\zeta) - \Phi(\zeta)\overline{\Psi(\zeta)}| < \varepsilon.$$

Proof. For each j , the one-variable Douglas–Rudin theorem [1] provides inner φ_j, ψ_j on $\mathbb{D}$ such that

$$\operatorname{ess\,sup}_{w \in \mathbb{T}} |g_j(w) - \varphi_j(w)\overline{\psi_j(w)}| < \varepsilon/N.$$

The character $\chi_{a_j}(\zeta) = \zeta^{a_j}$ pushes Haar measure on $\mathbb{T}^d$ forward to Haar measure on $\mathbb{T}$. Consequently, the exceptional null set in this estimate has null preimage under χ_{a_j} .

The functions $\varphi_j(z^{a_j})$ and $\psi_j(z^{a_j})$ are inner on $\mathbb{D}^d$. Indeed they are bounded and holomorphic; for almost every ζ , radial approach gives

$$\varphi_j((r\zeta)^{a_j}) = \varphi_j(r^{|a_j|}\zeta^{a_j}) \longrightarrow \varphi_j^*(\zeta^{a_j}),$$

whose modulus is one, and similarly for ψ_j . Thus

$$\Phi(z) = \prod_j \varphi_j(z^{a_j}), \quad \Psi(z) = \prod_j \psi_j(z^{a_j})$$

are polydisc inner. A telescoping product estimate for unit-modulus factors now gives

$$\|f - \Phi\overline{\Psi}\|_\infty \leq \sum_{j=1}^N \|g_j - (\varphi_j\overline{\psi_j})\|_\infty < \varepsilon.$$

□

Corollary 1. *Every coordinate-separable measurable phase $f(\zeta) = \prod_{k=1}^d g_k(\zeta_k)$ satisfies the full uniform conclusion. The same holds for every finite product of measurable phases of analytic monomial characters.*

5 Boundary of the result

Uniform phase quantization reduces the full problem to the signs $2\mathbf{1}_E - 1$ for arbitrary measurable $E \subset \mathbb{T}^d$. Theorem 1 cannot remove a set of small positive measure because the target norm is essential supremum. Theorem 2 depends on exact finite multiplicative separation through analytic characters, which a general set E need not possess. Eight distinct attempts at closing this gap—simple phase assembly, Hankel and Toeplitz obstructions, topology, slicing, operator-valued iteration, exceptional-set removal, and closure invariants—are documented in the associated attempt note; none settles the arbitrary sign.

A bounded novelty search on 11 August 2026 used the exact arXiv identifier and title, measurable-polydisc Douglas–Rudin phrases, rational-inner quotient density in finite L^r , and monomial-pullback phrases. It found the source and McDonald’s continuous result, but no later primary-source resolution of the measurable uniform question and no exact match for the two theorems above. Novelty confidence is moderate rather than definitive.

Human-review recommendation. Verify the citation-level formulation of McDonald’s theorem and the Haar pushforward step in Theorem 2. The proofs otherwise use only measure regularity, Urysohn separation, the classical one-variable theorem, and a telescoping inequality.

References

- [1] R. G. Douglas and W. Rudin, *Approximation by inner functions*, Pacific J. Math. **31** (1969), 313–320.
- [2] J. N. McDonald, *Approximation by quotients of rational inner functions*, Proc. Amer. Math. Soc. **74** (1979), 299–300.
- [3] P. Kumar, S. Rastogi, and D. Tripathi, *Douglas–Rudin Approximation theorem for operator-valued functions on the unit ball of $\mathbb{C}^d$* , arXiv:2403.16401, 2024.